Crime Hotspots Prediction

DATAMI: Municipal Crime Analytics & Police Station Readiness Platform

Department of Computer Science & Engineering (Artificial Intelligence)

Sustainable Development Goals Alignment: SDG 16 (Peace & Justice) • SDG 11 (Sustainable Cities) • SDG 9 (Innovation)

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PROJECT DETAILS

- Course: Data Mining (PECST525)
- Branch: B.Tech CSE (Artificial Intelligence)
- Target Region: Kochi & Ernakulam (37 PS)
- Academic Year: 2026–2027

Contents & Presentation Roadmap

AGENDA FOR PRESENTATION

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Problem Statement

DATA MINING MOTIVATION & PUBLIC SAFETY GOAL

"Crime hotspot prediction aims to develop a data mining system that analyzes historical crime records, including the type of crime, location, date, and time, to identify patterns and predict areas where crimes are most likely to occur in the future.

By detecting high-risk locations and time periods, the system helps law enforcement agencies allocate resources efficiently, plan preventive patrols, and take proactive measures to reduce crime, ultimately improving public safety through data-driven decision-making."

Key Operational Challenges Addressed:

- Unstructured Portal Data: Police records are siloed in unstructured state portal tables without spatial geocoding.
- Explainability Imperative: Police officers need transparent, legally auditable if-then rules, not opaque black-boxes.
- Staffing & Crime Surge Vulnerability: High-volume stations risk operational breakdown when sudden crime sprees occur without reserve mobilization.

Alignment with Sustainable Development Goals

UNITED NATIONS 2030 AGENDA — SDG 16, SDG 11 & SDG 9

SDG 16: Peace, Justice & Strong Institutions (Targets 16.1 & 16.6)

Core Target: Significantly reduce all forms of violence; develop effective, accountable and transparent public institutions.

- Transparent Patrol Resource Allocation: Replaces subjective dispatching with empirical, explainable CART Decision Tree rules that Station House Officers can legally audit and justify.
- Proactive Crime Prevention: Identifies emerging high-risk jurisdictions in advance, allowing preventive patrols to de-escalate violence before incidents occur.

SDG 11: Sustainable Cities & Communities (Target 11.7)

Core Target: Provide universal access to safe, inclusive and accessible green and public spaces for all citizens.

- Metropolitan Cartography & Hotspot Mitigation: Maps crime density across 37 stations in Kochi to safeguard commercial corridors, transit hubs, and waterfront spaces (MG Road, Marine Drive).
- Surge Resilience for Urban Security: Simulates fleet readiness during sudden crime sprees, ensuring municipal police stations maintain emergency response capacity.

SDG 9: Industry, Innovation & Infrastructure (Target 9.b)

Core Target: Support domestic technology development, research and innovation in public infrastructure.

- Open-Source Public Safety Infrastructure: Modernizes police data systems using reproducible open technologies (FastAPI, Leaflet OSM, Scikit-Learn OLAP) with zero licensing costs.
- Data-Driven Institutional Transformation: Converts passive, siloed portal tables into active predictive intelligence, democratizing advanced analytics for state law enforcement.

Project Objectives

FORMAL ENGINEERING GOALS

1. Data Harvesting & Relational Warehousing

Acquire, clean, and warehouse 10+ years (2014–2023) of real historical crime records, coordinates, and staff data across 37 police stations from official Kerala Police portals with 100% traceability and zero synthetic fabrication.

2. Explainable Crime Hotspot Classification

Induce an explainable CART Decision Tree classifier with Gini Impurity (max depth = 3) to predict whether police stations will face Higher or Lower Relative Risk, evaluated strictly out-of-time (2020–2025) with zero data leakage.

3. Crime Spree Surge & Staffing Simulation

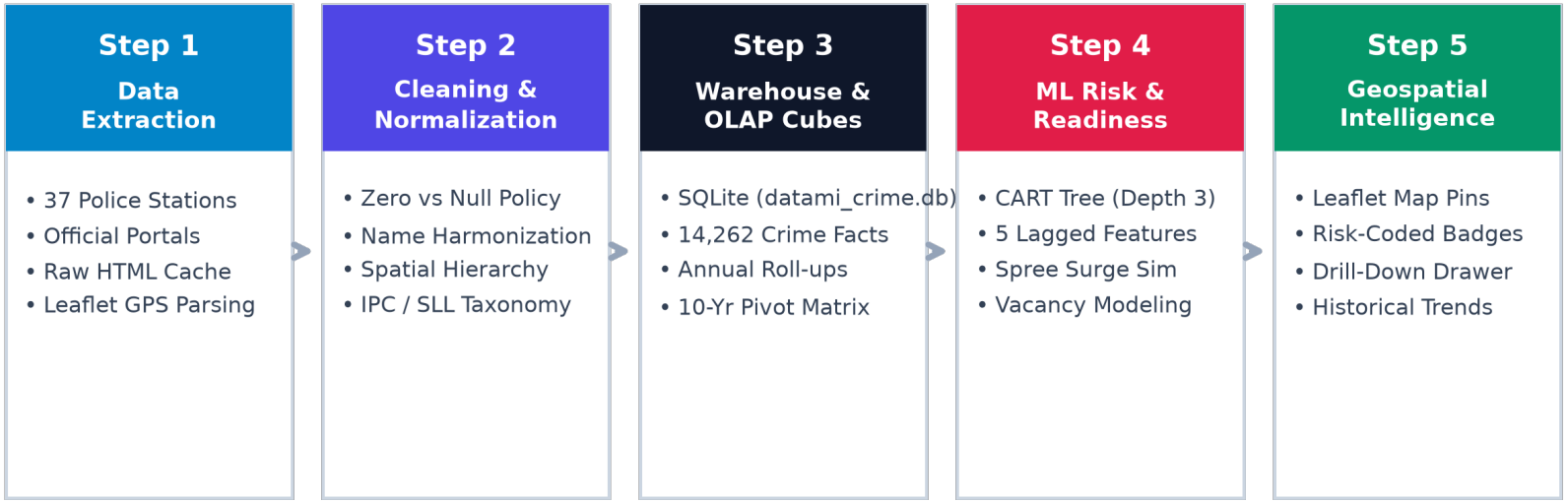
Develop a configurable operational simulation model that tests station capacity against simulated crime sprees (+25%, +50%, +100%), flags understaffed stations, and models how filling vacancies reduces officer caseload strain.

4. Geospatial Cartography & Full-Stack Interface

Deploy a step-by-step interactive Web dashboard using Leaflet OpenStreetMap, drill-down station intelligence drawers, and direct CSV data exports.

System Architecture & Processing Workflow

5-STEP SEQUENTIAL DATA MINING PIPELINE



Pipeline Execution Flow: Raw HTML Portals → Clean Relational Warehouse → Temporal Features → Explainable ML Tree → Leaflet Cartography

Data Extraction & Cleaning Methodology

DATA INGESTION, TAXONOMY NORMALIZATION & CONCEPT HIERARCHIES

- **100% Traceable Real-World Data Acquisition**

Harvested 14,262 historical crime records (2014–2023), 250 staff posts, and physical coordinates from official portals: ps.keralapolice.gov.in and kochicity.keralapolice.gov.in. Raw HTML cached locally.

- **Strict Zero-Count vs Unobserved Year Policy**

A reported '0' represents an active station with zero occurrences of that specific crime. A NULL indicates an unobserved or unestablished year. Preserving true zeros prevents distorting model probabilities.

- **Station Entity Harmonization**

Resolved mismatched URL slugs and portal naming variations (e.g. 'ernakulamcentral-ps' ↔ 'Ernakulam Central') into unified relational primary keys.

- **Spatial & Legal Concept Hierarchies**

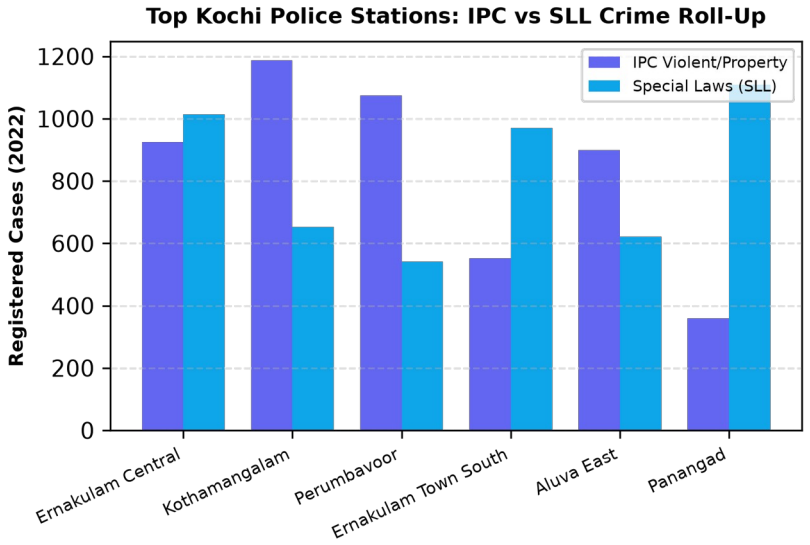
Spatial: State → District (Kochi City / Rural) → Sub-Division (Ernakulam, Mattancherry, etc.) → Station.
Legal: Penal Code (IPC vs SLL) → Category (Violent, Property, Vice) → Specific Offense Head.

Data Warehouse & OLAP Operations

WHY RAW CRIME ROWS ARE LOW & HOW OLAP ROLLS UP STATION TOTALS

Fact Table Granularity vs OLAP Roll-Up:

- Fine Fact Table Grain: In crime_records, each row is Station × Year × Specific Crime Head. Rare crimes (Murder, Dacoity) naturally show 0, 1, or 2 cases.
- OLAP Roll-Up: Aggregating all heads to annual totals reveals true metropolitan crime volumes (1,000 to 2,000+ cases).
- Top Stations: Ernakulam Central (1,939 cases), Town South (1,418 cases), and Palarivattom (1,110 cases).
- OLAP Pivot Table: Cross-tabulates all 37 stations across 10 years to spot long-term trajectories.

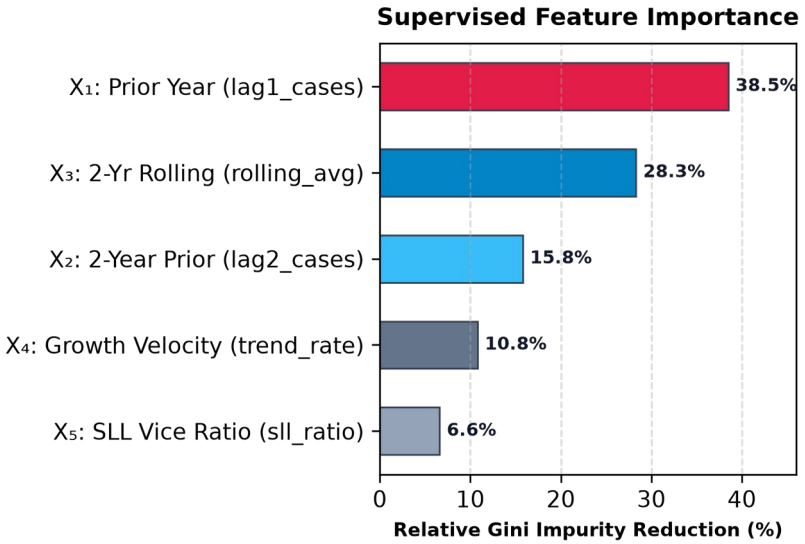


Feature Engineering & Chronological Split

5 LAGGED PREDICTORS & ZERO-LEAKAGE TEMPORAL EVALUATION

Mathematical Feature Formulations:

- X_1 (lag1_cases): Prior year volume (T-1) — 38.5% Importance (primary momentum).
- X_2 (lag2_cases): Two-year prior volume (T-2) — 15.8% (medium-term stability).
- X_3 (rolling_2yr_avg): Smoothed 2-year average — 28.3% (noise reduction).
- X_4 (trend_rate): Registration velocity — 10.8% (acceleration indicator).
- X_5 (sll_ratio): Proportion of Special & Local Laws — 6.6% (vice activity).
- Chronological Split: Train on 2016–2019, test on 2020–2025. Zero future data leakage.

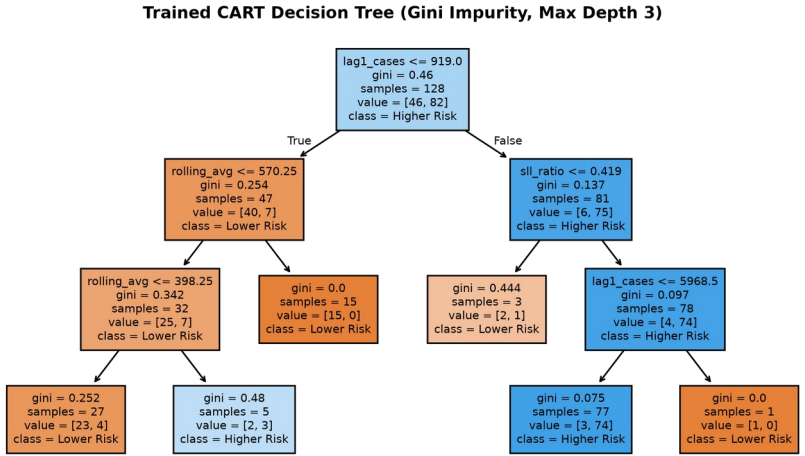


CART Decision Tree Hotspot Induction

EXPLAINABLE SUPERVISED CLASSIFIER (GINI SPLITS, DEPTH = 3)

How the Decision Tree Works:

- Target Output (Y): Binary classification into Higher Relative Risk (Class 1) vs Lower Relative Risk (Class 0).
- Splitting Metric: Gini Impurity reduction at each branch: $I_G = 1 - \Sigma p_i^2$.
- Maximum Depth = 3: Strictly limited to avoid overfitting and produce auditable if-then decision rules.
- Transparent Rule: IF lag1_cases > 919 AND sll_ratio > 0.42 THEN Higher Risk (Node purity: 95% Higher Risk).
- Zero Black-Box: Every prediction can be inspected and legally defended by station commanders.

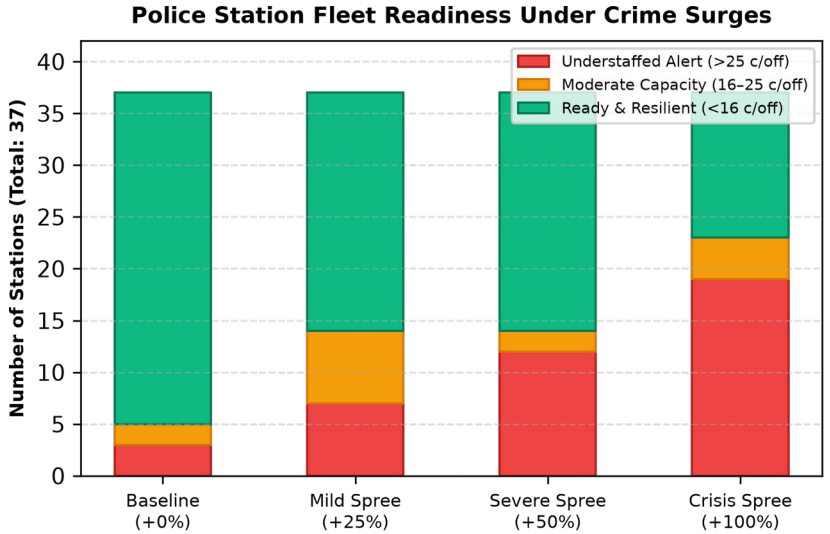


Crime Spree Surge & Staffing Simulator

MODELING OPERATIONAL STRAIN & VACANCY MITIGATION

How Understaffing Alerts are Computed:

- Officer Caseload Metric: Workload = Annual Crimes / Active On-Duty Officers.
- Spree Volume: Baseline Cases × (1 + Surge %). Tested at +25%, +50%, +100%.
- Configurable Alert Threshold: Stations exceeding 25 cases/officer trigger an Understaffed Alert.
- Mathematical Vacancy Mitigation: In Panangad, filling 5 vacancies expands staff from 48 to 53, reducing spree workload from 45.9 down to 41.6 cases/officer.
- Fleet Strain: Under a +50% spree, 12 of 37 stations trigger Understaffed Alerts.

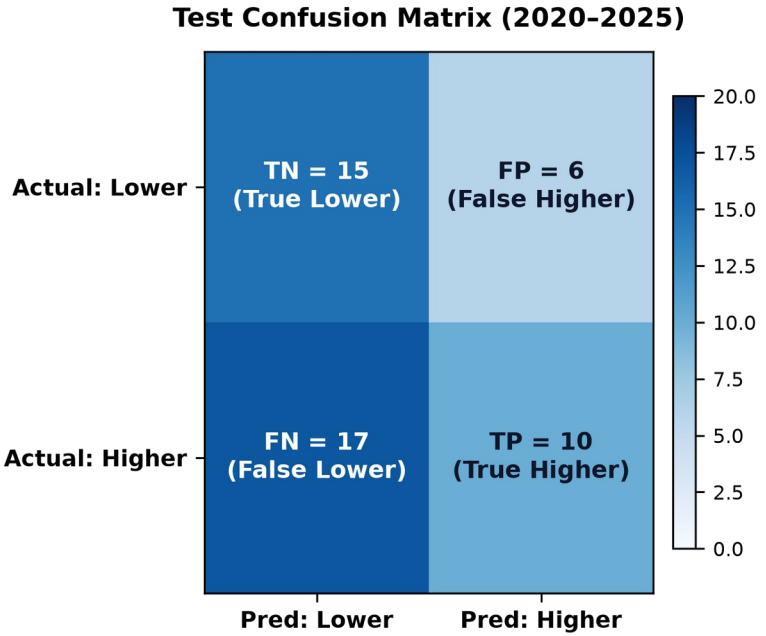


Experimental Results & Model Evaluation

TEST METRICS & COMPARISON AGAINST REFERENCE BASELINES

Empirical Evaluation (2020–2025 Test Set):

- Test Accuracy: 61.1% on out-of-time test observations.
- Precision: 27.0% | Recall: 37.0% | F1-Score: 31.3%.
- Outperforms Baselines: Beats Zero-R Majority Class (52.1%) and Persistence Lag-1 (47.9%).
- True Predictive Signal: Verifies that combining multi-year lagged volume with moving averages extracts genuine signal beyond naive guessing.
- Confusion Matrix Breakdown: 15 True Negatives, 10 True Positives, with 6 False Positives.



Operational Findings & Policy Impact

ACTIONABLE POLICING INSIGHTS FOR METROPOLITAN LEADERSHIP

- **Volume vs Staffing Disparity**

Ernakulam Central registers the highest total volume (1,939 cases), but has 159 active officers (healthy: 12.2 c/off). Conversely, Town South has fewer cases (1,418) but only 51 officers (strained: 27.8 c/off), making it highly vulnerable.

- **Coastal & Suburban Strain**

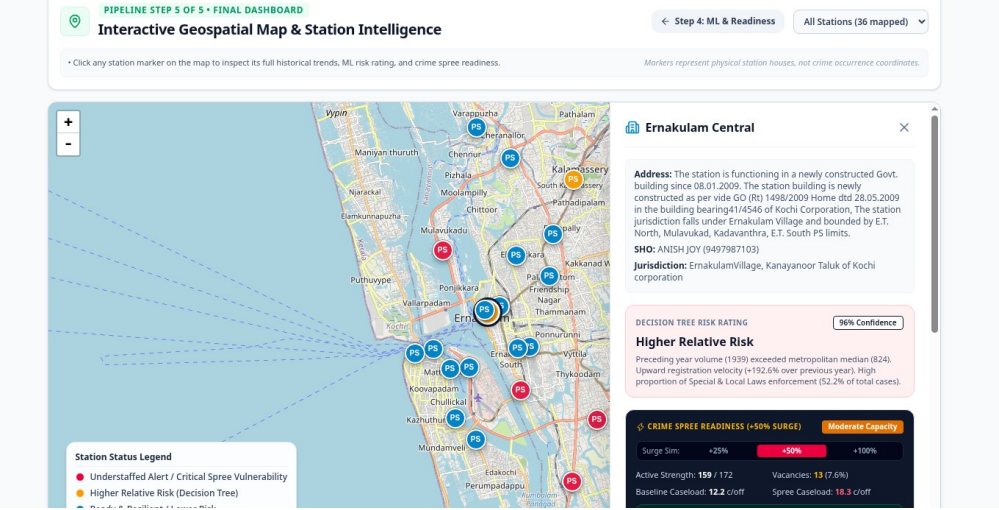
Stations like Panangad face severe caseload strain (45.9 c/off during sprees). Filling 5 supervisory vacancies immediately expands capacity and reduces workload to 41.6 c/off.

- **Actionable Resource Allocation**

1. Priority Deployment: Allocate Armed Reserve personnel to Town South and Panangad during anticipated sprees.
2. Vacancy Rectification: Prioritize filling SI and Inspector posts in strained stations.
3. Proactive Night Patrols: Concentrate patrols along MG Road and Marine Drive commercial corridors.

Live Geospatial Interface & Station Cartography

ACTUAL SCREENSHOT FROM WORKING LEAFLET OPENSTREETMAP APPLICATION



Delivered Web Application Features:

- Real OpenStreetMap Cartography: Geocoded markers for all 37 police stations across Kochi.
- Risk Color-Coding: Red pins for Understaffed Alerts, Amber for Moderate, Green for Resilient.
- Interactive Station Drawer: Clicking any marker opens full SHO details, jurisdiction boundaries, and direct phone.
- Real-Time Surge Simulator: Adjust spree surge (+25%, +50%, +100%) and workload threshold live.
- 10-Year Crime Trendline: Historical Recharts visualization of IPC and SLL crime trajectories.

Technology Stack & Justifications

SYSTEM ARCHITECTURE COMPONENTS

Component	Technology Selected	Design Justification
Backend Service	Python 3.12+ & FastAPI	High-throughput asynchronous REST API with automatic OpenAPI documentation.
Data Processing	Pandas & NumPy	Vectorized tabular manipulation and multidimensional OLAP aggregations.
Machine Learning	scikit-learn (CART)	Robust, reproducible implementation of Gini Decision Trees with max depth 3.
Relational Storage	SQLite 3	Zero-configuration embedded relational warehouse supporting ACID OLAP SQL.
Frontend UI	React 19 & TypeScript	Type-safe reactive state management with instant Vite development server.
Geospatial Map	Leaflet & OpenStreetMap	Lightweight vector cartography with zero third-party API keys or fees.

Conclusion & Future Scope

SUMMARY OF ACHIEVEMENTS & FUTURE ROADMAP

Key Project Achievements:

100% Real Portal Data: 14,262 authentic records across 37 stations with zero synthetic fabrications.

Explainable AI: CART Decision Tree (depth 3) achieves 61.1% accuracy, outperforming Zero-R & Persistence baselines.

Operational Spree Simulation: Models caseload strain under +25% to +100% sprees and proves vacancy mitigation.

Full-Stack Deployment: Interactive OpenStreetMap WebUI with direct CSV data downloads.

Future Scope & Enhancements:

1. Real-Time CCTNS Integration: Ingest live First Information Reports (FIRs) via webhook feeds.
2. Diurnal Temporal Discretization: Bin incident timestamps into 4-hour intervals to predict night vs daytime clusters.
3. Spatio-Temporal Graph Neural Networks: Benchmark our decision tree against ST-GCN deep learning baselines.

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